

Problem-Solution Fit

Date	10 JUNE 2026
Team ID	xxxxxx
Project Name	☐ Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas: Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Primary: Students in higher education or professional training tracks who need to demonstrate conceptual proficiency through speaking assignments. Secondary: Educators, teaching assistants, and institutional assessors looking to automate grading workflows for oral examinations and interview preparations	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Hardware Constraints: Students often possess low-to-mid tier client systems or low-cost built-in consumer laptop microphones prone to ambient noise. Computational Footprint: Lack of access to local high-end NVIDIA GPUs to run resource-heavy transformer configurations locally.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Viva-Voce Panels: <i>Pros:</i> High nuance understanding, situational flexibility. <i>Cons:</i> Not scalable, highly time-consuming, prone to human fatigue and subconscious evaluation bias.
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□ **Time Windows:** Limited time windows available during tight assignment submission schedules

- **Standard Written Text Exams:**
- *Pros:* Easy automated processing via keywords or multiplechoice interfaces.
- *Cons:* Completely misses a student's verbal articulation, pronunciation patterns, real-time explanation clarity, and authentic conceptual retrieval

2. PROBLEMS / PAINS

+ ITS FREQUENCY [PR] □

Subjective Evaluation Bias:

Traditional viva-voce and spoken assessments suffer from subconscious grader bias, leading to inconsistent marks across students.

(Frequency: Continuous during exam seasons).

- **Turnaround Delays:**

Manual listening and feedback reporting require immense time commitments, causing weeks of delay in releasing marks. *(Frequency: Regular post-assessment constraint).*

- **Lack of Actionable**

Insights: Traditional raw marks fail to pinpoint exact articulation flaws, such as unnecessary pauses or high filler-word density.

(Frequency: Every assignment lifecycle)

9. PROBLEM ROOT / CAUSE [RC]

- **Acoustic**

Suboptimization: Traditional software applications fail to map speech properties (such as silence thresholds or pause durations) directly to conceptual confidence levels.

- **Semantic Parsing**

Gaps: Basic keyword-matching programs cannot measure underlying conceptual intent or interpret synonyms accurately without standard NLP embedding pipelines.

7. BEHAVIOR

+ ITS INTENSITY [BE]

- **High Reliance on**

Memorization: When articulating under pressure, students frequently stutter, overuse filler words ("um," "like"), or exhibit extended silence gaps.

- **Panic**

Procrastination: Students frequently record assessments at the absolute last minute, driving intensive spikes in system submission queues *(Intensity: High)*

3. TRIGGERS TO ACT

[TR] □ **Scale Bottlenecks:** A sudden surge in class sizes or remote learning enrollments making manual viva-voce tests operationally impossible to scale.

- **Student Feedback Grievances:** Rising student complaints regarding opaque or perceived unfair qualitative grading criteria in spoken assignments.

- **Digital Transformation Mandates:** Institutional requirements to introduce objective, data-driven learning analytics frameworks.

10. YOUR SOLUTION

[SL]

Voice-Based Concept Understanding Analyser (VBCUA): An intelligent, automated evaluation platform built with a **Streamlit front-end**. It captures raw vocal explanations via an interface recorder, transcribes it using **OpenAI Whisper**, processes semantic proximity via **SentenceBERT**, extracts delivery flaws (fillers/pauses) via **Librosa**, and produces an automated **ReportLab PDF assessment breakdown** instantly.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

Interacting with Learning Management Systems (LMS), browsing institutional portals, or recording submissions via web browsers.

4. EMOTIONS

BEFORE / AFTER [EM] □

Before:

- *Students:* Anxious about subjective grading, nervous about performance consistency, and frustrated by vague feedback.
- *Educators:* Overwhelmed by backlogs of audio recordings to evaluate and worried about maintaining consistent rubric standards.
- **After:**
- *Students:* Relieved to receive instantaneous, data-validated

objective metrics, clear formatting summaries, and constructive analytics.

- *Educators:* Confident in automated, structured scoring checks that handle heavy grading workflows effortlessly.

OFFLINE

Participating in physical study sessions, reading reference textbooks, or presenting in standard classroom settings.